Divyam Ahuja

Systems & Software Engineer

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Systems & Software Engineer specializing in high-performance backend infrastructure, scalable automation platforms, and system-level performance tuning. Demonstrated expertise in architecting resilient distributed systems, complemented by a strong track record of contributions to massive open-source codebases.

Experience

Microsoft Noida
Software Engineer Mar 2025 - Present
Tech Stack: Python, TypeScript, Docker, Azure, React, Playwright

- Engineer and evaluate AI-powered experiences across multimodal models and agentic workflows for the Excel Copilot team.
- Owned the end-to-end Responsible AI (RAI) evaluation pipeline gating model releases; optimized execution latency by **50%** through asynchronous worker pools, intelligent request batching, and custom retry algorithms.
- Onboarded next-generation multimodal AI agents into the core evaluation platform, designing pluggable backend adapters, isolated container sandboxes (Docker), and concurrent distributed cloud runtimes.
- Designed and implemented a parallelized automation framework for synthetic data generation and evaluation; boosted throughput by **5x** and generated **70,000+** trajectories across **20,000+** enterprise workbooks.
- Served as on-call engineer driving infrastructure reliability across evaluation pipelines—triaging partner team issues, resolving live-site instabilities, and handling incident response to maintain **>95%** reliability.
- Integrated comprehensive telemetry pipelines and KQL aggregations processing **millions** of daily events, establishing single-pane system health dashboards and automated triage.

Advanced Micro Devices (AMD) Bangalore
System Software Designer Aug 2022 - Feb 2025
Tech Stack: C/C++, Java (OpenJDK), Linux perf, Python, Django, React, SQLite, Docker

- Engineered an OpenJDK performance monitoring and telemetry platform (Django, React, SQLite) to track nightly SPECjbb runs, enabling early detection and isolation of performance regressions.
- Automated systems to benchmark and tune Java nightly using machine learning models, achieving a **30%** reduction in manual effort and improved parameter accuracy.
- Applied advanced Java tuning techniques and profiled CPU/memory bottlenecks with Linux perf, yielding a **2%** throughput increase on SPECjbb benchmarks across High Performance Computing (HPC) servers.
- Refactored and containerized legacy benchmarking workflows with Docker for cross-platform execution, driving a **90%** reduction in deployment and environment errors.
- Awarded the **Spotlight Award** for exceptional contributions to automation, tuning, and nightly benchmark visualization.

Open Source

Chromium (Google Summer of Code) | C++, WebAssembly, WebAudio, WebWorkers
- Engineered a lock-free C++ ring buffer as a reference library for real-time audio streaming between WebAudio, WebAssembly, and WebWorkers, benchmarking multiple concurrency strategies.
- summerofcode.withgoogle.com

The Tor Project (Contributor) | Kotlin, Android SDK, Rust, JavaScript, VPNService
- Contributed across Tor Project client and web applications; engineered routing resilience and lifecycle management for the Tor VPN Android app, and developed enhancements for Tor Web platforms.
- gitlab.torproject.org/tpo

p5.js (Processing Foundation) (Contributor) | JavaScript, HTML5 Canvas, WebGL
- Contributed core library improvements and rendering fixes to p5.js, optimizing canvas event handling and cross-browser graphic performance.
- github.com/processing/p5.js

Skills

Programming Languages: C/C++, Python, Go (Golang), Java, Kotlin, TypeScript, Bash
Frameworks & Tools: PyTorch, Azure OpenAI, Android SDK / NDK, React/Next.js, Django/Flask, Node.js, gRPC, Emscripten, Playwright
Infrastructure & Databases: Kubernetes, Docker, Linux/POSIX, PostgreSQL, MongoDB, Redis, MySQL, SQLite, WebAssembly, WireGuard

Projects

Fastbin: A Code Sharing Webapp | HTMX, Go (Golang), gRPC, PostgreSQL, Docker, Kubernetes
- Built a high-performance code sharing backend in Go utilizing gRPC Protobuf microservices, PostgreSQL with connection pooling and optimized indexing, containerized with Docker and deployed on Kubernetes.
- git.divyam.dev/divyam/fastbin, fastbin.lab.divyam.dev

Homelab | Raspberry Pi 5, WireGuard, Kubernetes, Rancher, MetalLB
- Personal infrastructure hosting containerized services with WireGuard VPN tunneling, MetalLB load balancing, and Rancher-managed Kubernetes, enabling secure, high-availability external access.

Education

Deenbandhu Chhotu Ram University of Science and Technology — Haryana, India
Bachelor of Technology (B.Tech.) Computer Science — Aug 2018 - Aug 2022

Certifications

- MITx 6.86x — Machine Learning with Python: Linear Models to Deep Learning
- CS50AI: CS50's Introduction to Artificial Intelligence with Python